

Exploring Math  with EIGENMATH

Curvilinear grad curl div with EIGENMATH

$$\nabla \times \mathbf{F} := \text{curl } \mathbf{F} := \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{e}_1 & h_2 \mathbf{e}_2 & h_3 \mathbf{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix}$$

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0.1 Preface

In classical vector analysis with the operators *grad*, *div*, *curl* (=rot), *laplacian* plays within the Cartesian coordinate system in $\mathbb{R}^3$, the use of other coordinate systems leads to a multitude of adapted formulas that are difficult to memorize. Therefore, a set of four general 'curvilinear' formulas is presented here and applied using the CAS EigenMath. To use these formula's for the vector analysis operators it is recommended to remember and work with general 'curvilinear' coordinates and only afterwards specify the concrete coordinate system instead of memorize many (≥ 12) special formula's for special coordinate systems e.g. *cartesian*, *cylindrical*, *spherical*,... .

To this end, we first briefly address the familiar concepts of the *position vector* and *tangent vectors*. We then use these objects to formulate the expressions for the curvilinear differential operators. In the exercises, we demonstrate their specialization to typical coordinate systems used in physics and mathematics, such as cylindrical or spherical coordinates.

A more advanced description of the above differential operators can be provided using differential forms, as Andrew WINKLER gives so aptly in response to the question *How are a gradient, curl, divergence, Laplacian, etc. related to each other?*¹ on Quora:

They are all constructed from the 'exterior derivative', together with the 'dot product' and its associated 'Hodge duality'.

The exterior derivative of a function is a one form. You can use the dot product to convert that into a vector. That's the gradient.

The exterior derivative of a one form is a two form. You can use Hodge duality to convert it to a one form, and then the dot product to make it a vector. That's the curl.

The exterior derivative of a two form is a three form. You can use Hodge duality to convert it into a 0 form, that is a function. That's the divergence.

In three dimensions the exterior derivative of a three form is 0.

If you integrate the norm squared of the gradient of a function, you can integrate by parts to get an integral of the function times a differential operator applies to the function. That's the Laplacian.

But that would be the subject of another little book.²

- To implement the formulas of vector analysis, we use the CAS EIGENMATH: it is a small but well designed and powerful computer algebra system, that can be used to solve many problems in mathematics and the natural and engineering sciences. It is a personal resource for students, teachers and scientists. EIGENMATH is compact, capable and free.

¹A. Winkler, cf. relations

²I am writing a book with the title 'Differential Forms with REDUCE^{excalc}

- The EIGENMATH language allows to make the concepts of vector analysis executable, to coin math formulas and processes into 'run'able functions/procedures and therefore to allow own experiments. The results of one's thinking in the CAS language is immediately returned to the screen and helps to check the right understanding.

It is certainly possible nowadays to have vector analysis calculations automatically performed using AI (as we demonstrate in some instances, for example, for verification purposes). In my view, however, this text remains helpful for developing a mathematical understanding of the concepts and grasping them thoroughly and to gain valuable insight.

- EIGENMATH is particularly well-suited for handling the formulas of vector analysis, thanks to its exceptionally simple syntax for calculating derivatives (including partial derivatives) and the ubiquitous Jacobian determinant.

- For using a code script in the EIGENMATH language, no installation of any software is necessary, *everything runs directly online*: only run a script by clicking on a link like

 Run EIGENMATH script

and then pressing the [Run] button below on the right on the apps window: the calculation is made, allowing further free inputs from the user.

If you own a iMac, there is the option to install the app EIGENMATH free of charge and run the scripts by opening the corresponding `txt`-file into the app's workspace and press [Run], cf. [12].

I want to thank George WEIGT for the development of the free CAS EIGENMATH and EIGENMATH^{online} over nearly 20 years and for his friendly support with tips and hints while writing these notes.

Any feedback from the user is very welcome.

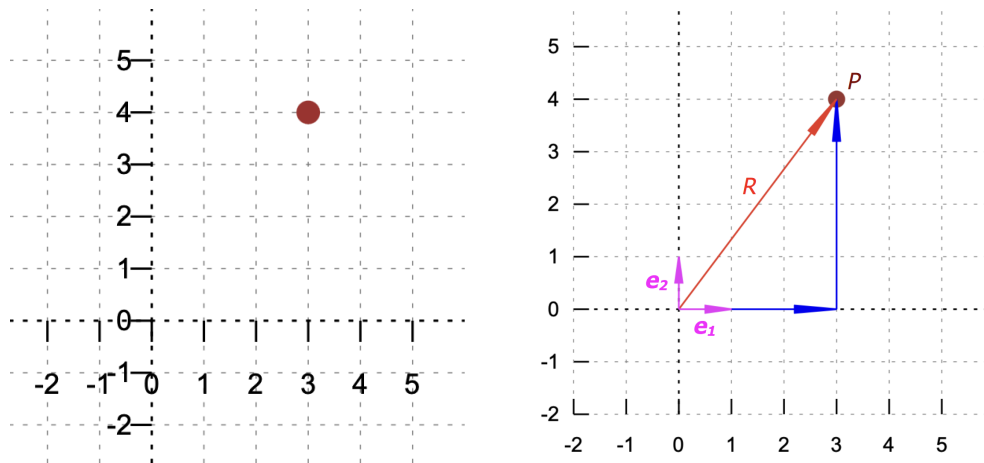
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July 2026

1 Position vector and Tangent vectors in $\mathbb{R}^3$

The concepts of VectorCalculus with general 'curvilinear' coordinates and the definitions of the operators *grad*, *curl*, *div* and *laplacian* relies largely on transformations between coordinate systems. Therefore the reader should be familiar with the basics of linear algebra (vector spaces, transformations) and analysis (partial derivative). FLEISCH [1] offers a very understandable introduction.

We recall the concepts of position vector and tangent vector. Therefore we use a short excerpt from [2, p.16–22].

1.1 Position vector in the Euclidian plane $\mathbb{R}^2$



left: point $P(3,4)$ with coordinates $x = 3$ and $y = 4$ in $\mathbb{R}^2$.

Figure 1: right: the standard basis $\mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $\mathbf{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, the position vector $R = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and the decomposition ' $\rightarrow\uparrow$ ' of R into $3 \cdot \mathbf{e}_1 + 4 \cdot \mathbf{e}_2$

In general, we associate to a point $P \in \mathbb{R}^3$ an triple (x_1, x_2, x_3) . The position vector R is then the vector $R := \vec{OP}$ from the origin of the coordinate system to the point P . $\square$

We begin with tree examples of tangent vectors to gain a visual mental image of this concept, which plays a central role beside the position vectors. Imagine a tangent vector as a vector that is tangent at a given point to a curve or a surface in $\mathbb{R}^3$.

1.2 Tangent vector on a Parabola

Imagine a mass point that travels along a parabolic path over time at a certain speed. If the mass point leaves the path at a certain position, it leaves the parabolic path *tangentially*. We denote the path by M ('manifold', i.e. a curve in plane or a surface in space), the position vector of the point at time t by $R(t)$ and the velocity by v .

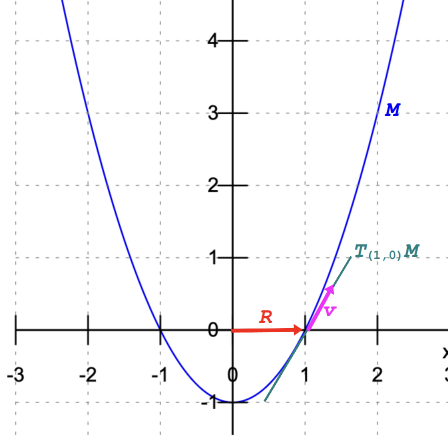


Figure 2:
 M : parabola $\{(x, y) \in \mathbb{R}^2 | y = x^2 - 1\}$, curve of $R(t) := (t, t^2 - 1)$.
 R : the position vector $R := R(1) = (0, 1)$.
 v : the velocity vector $R'(t) = (1, 2t)$.
 $T_{(0,1)}M$: tangent line at point $P := R(1)$ on M .

Then the velocity is defined as follows:

$$\underset{\text{Math}}{\mathbf{v}} := \frac{\partial R}{\partial t} = \underset{\text{Eigenmath}}{d(R(t), t)} \quad (1.1)$$

- We calculate the velocity v in Fig.2 first mathematically.

Let $R : I \rightarrow \mathbb{R}^2$ with $t \mapsto (t, t^2)$ be differentiable on the open interval $I \subset \mathbb{R}$.

Then $R'(t) = \frac{\partial}{\partial t}(t, t^2) = (1, 2t)$ and $R'(1) = (1, 2) = v$ is the *tangential vector* at the time $t = 1$ alias at the point $P = (0, 1)$.

- Then we calculate v using EIGENMATH:

```

R(t) = (t,t^2-1)      --* parametrization of parabola
R(t)
v = d(R(t),t)         --* velocity
v
Tp = eval(v,t,1)      --* tangent vector at [0,1]
Tp

```

 Run EIGENMATH script

EIGENMATH output:

$$\begin{aligned} & \begin{bmatrix} t \\ t^2 - 1 \end{bmatrix} \\ v &= \begin{bmatrix} 1 \\ 2t \end{bmatrix} \\ T_p &= \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{aligned}$$

Exercise 1. (tangent vector on a circle)

The unit circle is a 1-dimensional manifold ('curve') in $\mathbb{R}^2$.

We recommend to describe this circle $S^1 := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$ via a parametrization using the position vector $R(t) := (\cos(t), \sin(t))$.

Calculate the tangent vector at $P = R(45^\circ)$.

You should reproduce the following scene:

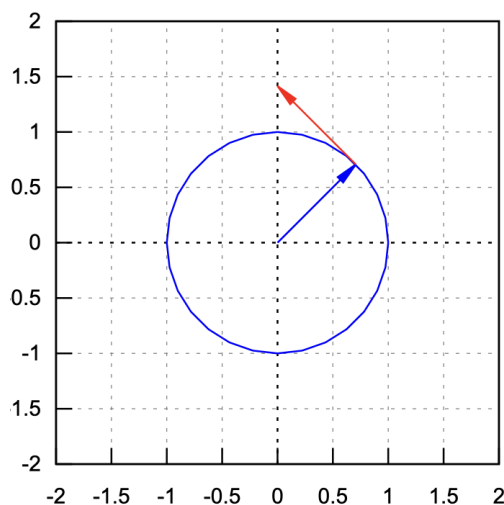


Figure 3: $R(\pi/4)$: position vector at time $t = \pi/4$ pointing on S^1 .
 $\partial_t R$: tangent vector $R'(\pi/4)$ at $R(\pi/4) \in S^1$.

1.3 Tangent vectors on a Sphere

The unit sphere S^2 is a 2-dimensional manifold ('surface') in the surrounding space $\mathbb{R}^3$. Let's describe the sphere $S^2 := \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\}$ by means of a parametrization using the position vector $R(\theta, \phi)$.

Let $R : U \rightarrow \mathbb{R}^3$ be defined by

$$(\theta, \phi) \mapsto (\cos(\theta) \cos(\phi), \cos(\theta) \sin(\phi), \sin(\theta)) \quad (1.2)$$

and differentiable on an open subset $U \subset \mathbb{R}^2$ with $(\theta, \phi) \in U$.

Then we have two tangent vectors: $\mathbf{Z}_1 := \frac{\partial R}{\partial \theta}$ and $\mathbf{Z}_2 := \frac{\partial R}{\partial \phi}$ are the *tangential vectors* at the point $R(\theta, \phi) \in S^2$ with coordinates (θ, ϕ) , which span the tangential plane $T_P M$.

Task:

- Calculate the tangent vectors at $P = R(\theta, \phi)$.
- Calculate the tangent vectors at $P = R(45^\circ, 0)$.

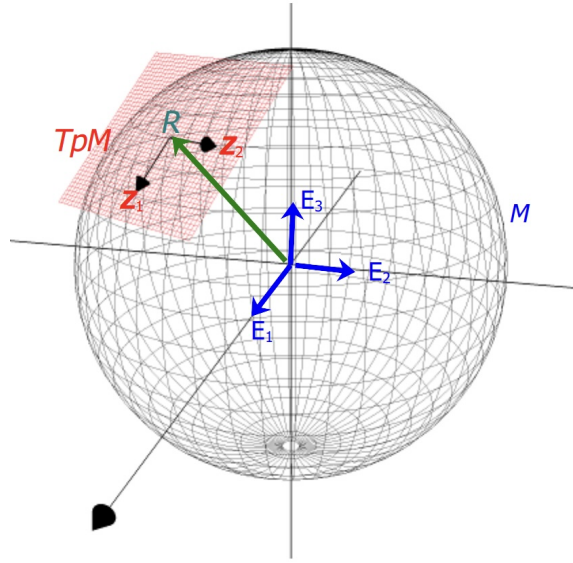


Figure 4: (E_1, E_2, E_3) : canonical Euclidian basis in $\mathbb{R}^3$.
 (Z_1, Z_2) : basis of tangent plane $T_P S^2$ at $R(\theta, \phi) \in S^2$.
 R : the position vector from the origin O to $R(\theta, \phi)$.

Solution:

1st mathematically.

a:

$$\begin{aligned}\mathbf{Z}_1 &= \frac{\partial R}{\partial \theta} \\ &= \frac{\partial}{\partial \theta}(\cos(\theta) \cos(\phi), \cos(\theta) \sin(\phi), \sin(\theta)) \\ &= (-\cos(\phi) \sin(\theta), -\sin(\phi) \sin(\theta), \cos(\theta))\end{aligned}$$

$$\begin{aligned}\mathbf{Z}_2 &= \frac{\partial R}{\partial \phi} \\ &= \frac{\partial}{\partial \phi}(\cos(\theta) \cos(\phi), \cos(\theta) \sin(\phi), \sin(\theta)) \\ &= (-\sin(\phi) \cos(\theta), \cos(\phi) \cos(\theta), 0)\end{aligned}$$

b:

$$\begin{aligned}\mathbf{Z}_1 \mid \left(\frac{\pi}{4}, 0\right) &= \left(\frac{-1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \\ \mathbf{Z}_2 \mid \left(\frac{\pi}{4}, 0\right) &= \left(0, \frac{1}{\sqrt{2}}, 0\right)\end{aligned}$$

2nd solution with EIGENMATH.

```
R(theta,phi) = (cos(theta)*cos(phi), cos(theta)*sin(phi), sin(theta))

Z1 = d( R(theta,phi), theta)
Z1
Z2 = d( R(theta,phi), phi)
Z2
eval(Z1, theta,pi/4, phi,0)
eval(Z2, theta,pi/4, phi,0)
```

 Run EIGENMATH script

EIGENMATH output:

$$\begin{aligned}Z_1 &= \begin{bmatrix} -\cos(\phi) \sin(\theta) \\ -\sin(\phi) \sin(\theta) \\ \cos(\theta) \end{bmatrix} & \begin{bmatrix} -\frac{1}{2^{1/2}} \\ 0 \\ \frac{1}{2^{1/2}} \end{bmatrix} \\ Z_2 &= \begin{bmatrix} -\cos(\theta) \sin(\phi) \\ \cos(\phi) \cos(\theta) \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ \frac{1}{2^{1/2}} \\ 0 \end{bmatrix} \quad \text{at } (\pi/2, 0) : \end{aligned}$$

Exercise 2. (tangent vectors on a cylinder)

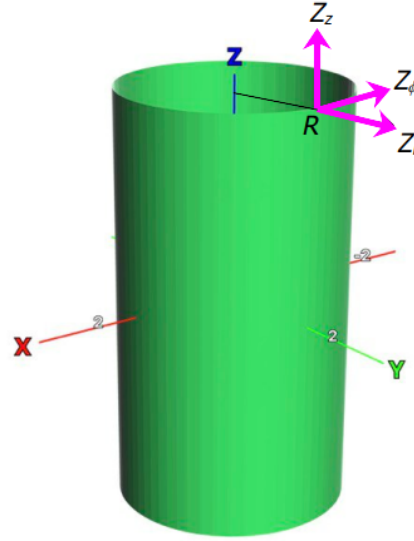
The cylinder of radius R is a 2-dimensional manifold ('surface') in $\mathbb{R}^3$.

Align a background Cartesian grid (x, y, z) with the cylindrical coordinates (r, θ, z) .

The position vector R , pointing to a position on the cylinder, is then given by

$$R(r, \theta, z) = (r \cos(\theta), r \sin(\theta), z) \quad (1.3)$$

Calculate the tangent vectors at $P = R(1, 45^\circ, 2)$.



Cylindrical coordinates: changing basis $\mathbf{e}_r = \mathbf{Z}_r, \mathbf{e}_\theta = \mathbf{Z}_\theta, \mathbf{e}_z = \mathbf{Z}_z$.

$R(r = 1, \theta = \pi/2, z = 2)$: position vector in $\mathbb{R}^3$.

Figure 5: $\partial_r = \mathbf{Z}_r$: 1st basis vector $\mathbf{Z}_r$ at R , change with the point.

$\partial_\theta = \mathbf{Z}_\theta$: 2nd basis vector $\mathbf{Z}_\theta$ at R , changing.

$\partial_z = \mathbf{Z}_z$: 3rd basis vector $\mathbf{Z}_z$ at R , changing.

Solution:

The basis vectors $\mathbf{Z}_i$ are the tangential vectors on the three coordinate lines, i.e. the basis is obtained by partial differentiation of R with respect to r, θ and z :

$$\mathbf{Z}_r = \frac{\partial R}{\partial r} \quad \mathbf{Z}_\theta = \frac{\partial R}{\partial \theta} \quad \mathbf{Z}_z = \frac{\partial R}{\partial z}$$

Remember: at each point $R(r, \theta, z)$ we get *another* 'new' basis $(\mathbf{Z}_r, \mathbf{Z}_\theta, \mathbf{Z}_z)$!

We skip the mathematical calculation using Calculus (left to the reader) and do the Mathematics directly in EIGENMATH.

```
# EIGENMATH
-- we introduce cylindrical coordinates in R^2 by

R(r,theta,z) = (r cos(theta), r sin(theta), z)
R

-- the basis in cylindrical coordinates
Z = d(R,(r,theta,z)) -- basis vectors are in the columns
Z = transpose(Z)      -- therefore we have to transpose
Z
Z[1]                  -- 1.st basis vector
eval(Z[1], r,1, theta,pi/4, z,2)
```

 Run EIGENMATH script

EIGENMATH output:

$$R = \begin{bmatrix} r \cos(\theta) \\ r \sin(\theta) \\ z \end{bmatrix}$$

$$Z = \begin{bmatrix} \cos(\theta) & \sin(\theta) & 0 \\ -r \sin(\theta) & r \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \cos(\theta) \\ \sin(\theta) \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{2^{1/2}} \\ \frac{1}{2^{1/2}} \\ 0 \end{bmatrix}$$

2 grad, div, curl, laplacian in general coordinates

To use formula's for the vector analysis 'operators' *grad*, *div*, *curl* (=rot), *laplacian* it is recommended to remember and work with general coordinates u for the definitions and afterwards specify the concrete coordinate system instead of memorize many (≥ 12) special formula's for special coordinate systems e.g. *cartesian*, *cylindrical*, *spherical*, ..see (1')..(4'), (1^C)..(4^C) and (1^S)..(4^S).

Theorem (Vector Analysis in $\mathbb{R}^3$ with general curvilinear coordinates u)

Let $u = (u_1, u_2, u_3)$ be a general ('curvilinear') orthogonal coordinate system.

Let $\mathbf{F} = (F_1, F_2, F_3): D \subset \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a differentiable vector function with coord's u .

Let $f(u_1, u_2, u_3): D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ be a differentiable "scalar" function with coordinates u .

Let $R: (x, y, z) \mapsto (u_1, u_2, u_3)$ be transformation between (x, y, z) and (u_1, u_2, u_3) coord's.

Let $\mathbf{U} := \left(\frac{\partial \mathbf{R}}{\partial u}\right) = (U_1, U_2, U_3)$ be the *Jacobian* matrix of R with columns U_i .

Let $h = (h_1, h_2, h_3)$ with $h_i := \sqrt{U_i \bullet U_i}$ be the *scale factors* of the transformation R .

Let $\mathbf{e}_i := \frac{1}{h_i} \mathbf{U}_i = \frac{1}{h_i} \frac{\partial \mathbf{R}}{\partial u_i}$ be the unit vector in direction U_i .

Then we define the 'symbolic' operators $\nabla f, \nabla \cdot \mathbf{F}, \nabla \times \mathbf{F}, \Delta f$ alias *grad*, *div*, *curl* (=rot), *laplacian* through the following explicit formular's on the right hand side:

$$\nabla f(u_1, u_2, u_3) := \text{grad}(f(u_1, u_2, u_3)) := \frac{1}{h_1} \frac{\partial f}{\partial u_1} \mathbf{e}_1 + \frac{1}{h_2} \frac{\partial f}{\partial u_2} \mathbf{e}_2 + \frac{1}{h_3} \frac{\partial f}{\partial u_3} \mathbf{e}_3 \quad (2.1)$$

$$\nabla \cdot \mathbf{F} := \text{div} \mathbf{F} := \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} (h_2 h_3 F_1) + \frac{\partial}{\partial u_2} (h_3 h_1 F_2) + \frac{\partial}{\partial u_3} (h_1 h_2 F_3) \right] \quad (2.2)$$

$$\nabla \times \mathbf{F} := \text{curl} \mathbf{F} := \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{e}_1 & h_2 \mathbf{e}_2 & h_3 \mathbf{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix} \quad (2.3)$$

$$\begin{aligned} &:= \frac{1}{h_2 h_3} \left(\frac{\partial(h_3 F_3)}{\partial u_2} - \frac{\partial(h_2 F_2)}{\partial u_3} \right) \mathbf{e}_1 \\ &+ \frac{1}{h_3 h_1} \left(\frac{\partial(h_1 F_1)}{\partial u_3} - \frac{\partial(h_3 F_3)}{\partial u_1} \right) \mathbf{e}_2 + \frac{1}{h_1 h_2} \left(\frac{\partial(h_2 F_2)}{\partial u_1} - \frac{\partial(h_1 F_1)}{\partial u_2} \right) \mathbf{e}_3 \end{aligned} \quad (2.4)$$

$$\Delta f := \text{laplacian}(f(u_1, u_2, u_3)) \quad (2.5)$$

$$:= \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial u_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial u_1} \right) + \frac{\partial}{\partial u_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial u_2} \right) + \frac{\partial}{\partial u_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial u_3} \right) \right]$$

Here is the implementation of the vector analysis 'operators' *grad*, *div*, *curl* (=rot), *laplacian* in

Eigenmath (Implementation of the Vector Analysis operators)

```
# EIGENMATH
" DEFINITIONS for curvilinear coordinates"
-- u : curvilinear 'general' coordinates
-- f : scalar function
-- F : vector function
-- h : scale factors
-- Jac : Jacobi determinant

F = (f1(),f2(),f3())                      ----* (1)

Grad(f) = ( d(f,u[1])/h[1], d(f,u[2])/h[2], d(f,u[3])/h[3] )

Div(F) = 1/Jac * ( d(Jac F[1]/h[1],u[1]) +
                  d(Jac F[2]/h[2],u[2]) +
                  d(Jac F[3]/h[3],u[3]))

Curl(F)= 1/Jac *(h[1]*(d(h[3]*F[3],u[2]) - d(h[2]*F[2],u[3])),
                h[2]*(d(h[1]*F[1],u[3]) - d(h[3]*F[3],u[1])),
                h[3]*(d(h[2]*F[2],u[1]) - d(h[1]*F[1],u[2])))

Laplacian(f)= 1/Jac *( d(h[2]*h[3]/h[1]*d(f,u[1]),u[1]) +
                      d(h[3]*h[1]/h[2]*d(f,u[2]),u[2]) +
                      d(h[1]*h[2]/h[3]*d(f,u[3]),u[3]) )
```

 Run EIGENMATH script

Comment. This is the translation of the formulae (2.1) to (2.5) to EIGENMATH. These definitions are used as prelude in concrete calculations, which is demonstrated in the following three examples §3.1 to §3.3. In (1) F is a so-called *template function* with just a name and an argument list, cf. [11, §14]. In the definition of $Div(F)$ we use the space as multiplication sign.

The terms for F , $Grad(F)$ and $Curl(F)$ calculates *only* the *coordinates* of these objects in the system in question. So we have to expand the result in the actual basis if wanted, i.e. $F = (x^2, y, z - 1) \equiv x^2 \cdot I + y \cdot J + (z - 1) \cdot K$ and I, J, K as cartesian basis in $\mathbb{R}^3$.

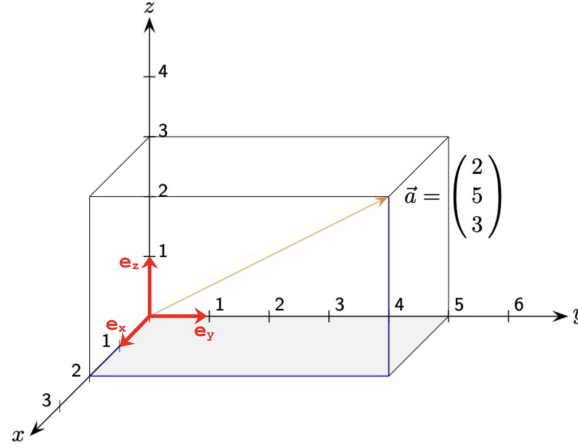
To solve concrete vector analysis problems in general coordinates one only have to submit the vector u and the position vector R , which is then used to define the Jacobian determinant Jac , the tangential basis U and the scale factors h .

The use of these EIGENMATH functions for cartesian, cylindrical and spherical coordinate systems is demonstrated in §3.1, §3.2 and §3.3.

3 Exercises

Verify using EIGENMATH.

3.1 rectangular cartesian coordinates



Cartesian coordinates: not changing, fix basis $\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z$ alias $\mathbf{i}, \mathbf{j}, \mathbf{k}$ alias $\mathbf{I}, \mathbf{J}, \mathbf{K}$.
We have: $\vec{a} = 2\mathbf{e}_x + 5\mathbf{e}_y + 3\mathbf{e}_z \equiv 2\mathbf{i} + 5\mathbf{j} + 3\mathbf{k}$.

Get the usual standard expressions for the 4 operators using **rectangular cartesian coordinates** (x, y, z) for (u_1, u_2, u_3) in the formular's (1)..(4):

$$(1') \quad \nabla f = \text{grad}(f) = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} \equiv \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

$$(2') \quad \nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$(3') \quad \nabla \times \mathbf{F} = \text{curl}(\mathbf{F}) = (F_{3,y} - F_{2,z})\mathbf{i} + (F_{1,z} - F_{3,x})\mathbf{j} + (F_{2,x} - F_{1,y})\mathbf{k} \quad \dagger^3$$

$$= \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) \mathbf{i} + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) \mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) \mathbf{k}$$

$$(4') \quad \Delta f \equiv \nabla^2 f \equiv \text{Laplacian}(f) := \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

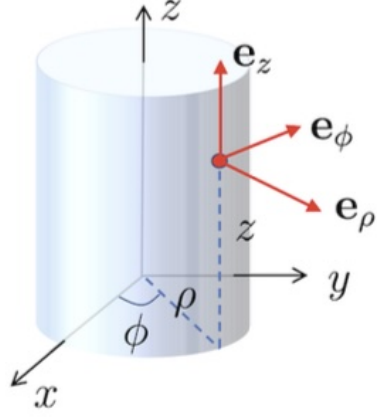
- a. If $f(x, y, z) = x^2 y z^3$ and $\mathbf{F} = xz\mathbf{i} - y^2\mathbf{j} + 2x^2 y \mathbf{k}$,
find $\nabla f, \nabla \cdot \mathbf{F}, \nabla \times \mathbf{F}, \text{div}(f\mathbf{F}), \text{curl}(f\mathbf{F}), \Delta f$.

 Run EIGENMATH script

- b. Check a. by hand.

³this is a usual compact short notation for the partial derivative: $\frac{\partial F_3}{\partial y} \equiv F_{3,y}$

3.2 cylindrical coordinates



Cylindrical coordinates: changing basis $\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{e}_z$, cf. [8].

Get the following expressions for the 4 operators using **cylindrical coordinates** (r, φ, z) for (u_1, u_2, u_3) in the formular's (1)..(4):

$$(1^C) \quad \begin{aligned} \nabla f &= \text{grad}(f) = f_{,r} \mathbf{e}_r + \frac{f_{,\varphi}}{r} \mathbf{e}_\varphi + f_{,z} \mathbf{e}_z \\ \nabla f(r, \varphi, z) &= \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \varphi} \mathbf{e}_\varphi + \frac{\partial f}{\partial z} \mathbf{e}_z \end{aligned}$$

$$(2^C) \quad \begin{aligned} \nabla \cdot \mathbf{F} &= \text{div } \mathbf{F} = \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} F_{\varphi,\varphi} + F_{z,z} \\ &= \frac{1}{r} \frac{\partial}{\partial r} (r F_r) + \frac{1}{r} \frac{\partial F_\varphi}{\partial \varphi} + \frac{\partial F_z}{\partial z} \end{aligned}$$

$$(3^C) \quad \nabla \times \mathbf{F} = \left[\frac{1}{r} \frac{\partial F_z}{\partial \varphi} - \frac{\partial F_\varphi}{\partial z} \right] \mathbf{e}_r + \left[\frac{\partial F_r}{\partial z} - \frac{\partial F_z}{\partial r} \right] \mathbf{e}_\varphi + \frac{1}{r} \left[\frac{\partial}{\partial r} (r \cdot F_\varphi) - \frac{\partial F_r}{\partial \varphi} \right] \mathbf{e}_z$$

$$(4^C) \quad \Delta f = \nabla^2 f = \text{Laplacian}(f) = \frac{f_{,r}}{r} + f_{,rr} + \frac{f_{,\varphi\varphi}}{r^2} + f_{,zz}$$

- a. If $f(r, \varphi, z) = r^2 z^3$ and $\mathbf{F} = z \mathbf{e}_r - \varphi^2 \mathbf{e}_\varphi + 2r^2 z \mathbf{e}_z$ find $\nabla f, \nabla \cdot \mathbf{F}, \nabla \times \mathbf{F}, \text{div}(f\mathbf{F}), \text{curl}(f\mathbf{F}), \Delta f$.

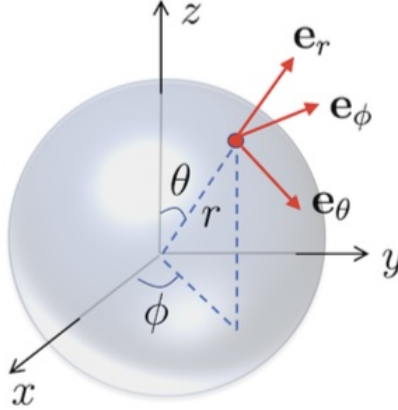
 Run EIGENMATH script

- b. Check a. by hand.

- c. With AI (google): ask the question

- calculate curl of $F(r, \text{phi}, z) = (z, -\text{phi}^2, 2 * r^2 * z)$ in cylindrical coordinates and check our solution in a.

3.3 spherical coordinates



Spherical coordinates: changing basis $\mathbf{e}_r, \mathbf{e}_\varphi, \mathbf{e}_\theta$, cf. [8].

Get the following expressions for the 4 operators using **spherical coordinates** (r, θ, φ) for (u_1, u_2, u_3) in the formular's (1)..(4):

$$(1^S) \quad \begin{aligned} \nabla f &= \text{grad}(f) = f_{,r} \mathbf{e}_r + \frac{f_{,\theta}}{r} \mathbf{e}_\theta + \frac{f_{,\varphi}}{r \sin(\theta)} \mathbf{e}_\varphi \\ \nabla f(r, \theta, \varphi) &= \frac{\partial f}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \mathbf{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \varphi} \mathbf{e}_\varphi \end{aligned}$$

$$(2^S) \quad \nabla \cdot \mathbf{F} = \text{div } \mathbf{F} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 F_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta F_\theta) + \frac{1}{r \sin \theta} \frac{\partial F_\varphi}{\partial \varphi}$$

$$(3^S) \quad \nabla \times \mathbf{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (F_\varphi \sin \theta) - \frac{\partial F_\theta}{\partial \varphi} \right] \mathbf{e}_r + \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial F_r}{\partial \varphi} - \frac{\partial}{\partial r} (r F_\varphi) \right] \mathbf{e}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} (r F_\theta) - \frac{\partial F_r}{\partial \theta} \right] \mathbf{e}_\varphi$$

$$(4^S) \quad \begin{aligned} \Delta f &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2 f}{\partial \varphi^2} \\ &= \frac{1}{r} \frac{\partial^2}{\partial r^2} (r f) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \varphi^2} \end{aligned}$$

- a. If $f(r, \theta, \varphi) = \frac{1}{2}(2 \cos \theta + 3 \sin^2 \theta \cos \varphi)$ and $\mathbf{F} = \mathbf{e}_r + \frac{2}{\theta} \mathbf{e}_\theta + \varphi r \sin(\theta) \mathbf{e}_\varphi$ find $\nabla f, \nabla \cdot \mathbf{F}, \nabla \times \mathbf{F}, \text{div}(f\mathbf{F}), \text{curl}(f\mathbf{F}), \Delta f$ in *spherical coordinates*.

 Run EIGENMATH script

b. Check a. by hand.

c. Check the results with AI (e.g Google) by asking appropriate questions.

3.4 Proofs.

Proof the formular's (1)..(4). See: [7, p. 137], [8, p. 62 ff].

- (1) Grad: see [9, p. 22], [7, p. 148], [8, p. 64], [4, §6.9]
- (2) Divergence: [9, p. 30], [7, p. 150], [8, p. 64], [4, §6.10]
- (3) Curl: [9, p. 31], [7, p. 150], [4, §6.11]
- (4) Laplacian: [9, p. 23], [7, p. 151], [4, §6.12]

3.5 Miscellaneous Exercises

Remark. The use of the EIGENMATH functions for the curvilinear expressed vector calculus operators `grad`, `div`, `curl`, `laplacian` in cartesian, cylindrical and spherical coordinate systems was demonstrated, derived and explained and in §3.1, §3.2 and §3.3. In the following exercises you may use predefined options sets for the use in the three prototypical coordinate systems. Therefore use one of the user-defined EIGENMATH commands.

`cartesian` `cylindrical` `spherical`

to invoke the adapted setting for the four curvilinear functions. You may define other user-defined options for other coordinate systems using these patterns.
See e.g. the following exercise 4.

Exercise 3. (conversions)

- Express the cylindrical coordinates $(2, \frac{2\pi}{3}, -2)$ as rectangular coordinates.
- Convert the cartesian coordinates $(3, 4, 2)$ to cylindrical coordinates.
- Convert the spherical coordinates $(16, \pi/3, \pi/6)$ to cylindrical coordinates.

 Run EIGENMATH script

Exercise 4. (vector operators in cylindrical coordinates)

- Answer the following questions from youTube: Physics - Advanced E&M.
Use EIGENMATH.

Gradient in Cylindrical Coordinates

Given: $f = \rho \sin \phi$

Find: $\nabla f = ?$

$$\nabla f = \frac{\partial f}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{\partial f}{\partial z} \hat{z}$$

Divergence in Cylindrical Coordinates

$$\nabla \cdot \vec{V} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho V_{\rho}) + \frac{1}{\phi} \frac{\partial V_{\phi}}{\partial \phi} + \frac{\partial V_z}{\partial z}$$

Given:

$$\vec{V} = \rho(2 + \sin^2 \phi) \hat{\rho} + \rho \sin \phi \cos \phi \hat{\phi} + 3z \hat{z}$$

Find: $\nabla \cdot \vec{V} = ?$

 Run EIGENMATH script

- Check your solutions using the free CAS WEB REDUCE, cf. [10].

 Run WEBREDUCE.

Exercise 5. (Laplacian in cylindrical coordinates)

Do the example in EIGENMATH using curvilinear coordinates, cf. youTube Physics.

LAPLACIAN (CYLINDRICAL COORDINATES)

$\nabla^2 = \frac{1}{r} \cdot \frac{\partial}{\partial r} \cdot \left(r \cdot \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \cdot \frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}$

EXAMPLE

$f(r, \theta, z) = r^2 \cdot \sin \theta + 2z$

$\frac{\partial f}{\partial r} = 2r \sin \theta$

$\frac{\partial f}{\partial \theta} = r^2 \cos \theta$

$\frac{\partial^2 f}{\partial \theta^2} = -r^2 \sin \theta$

$\frac{\partial f}{\partial z} = 2$

$\frac{\partial^2 f}{\partial z^2} = 0$

$\nabla^2 f(r, \theta, z) = \frac{1}{r} \cdot \frac{\partial}{\partial r} (r \cdot 2r \sin \theta) + \frac{1}{r^2} \cdot (-r^2 \sin \theta) + 0$

$\nabla^2 f(r, \theta, z) = \frac{1}{r} \cdot 4 \cdot r \cdot \sin \theta - \sin \theta$

$\nabla^2 f(r, \theta, z) = 3 \cdot \sin \theta$

Input/Output Display

Web REDUCE (6745), 2-Apr-2024 ...

```
load_package avector;
coordinates r,theta,z;
scalefactors(1,r,1);
f := r^2*sin(theta) + 2 z;
delsq(f);

(r theta z)
```

$f := \sin(\theta) r^2 + 2 z$

$3 \sin(\theta)$

Cylindrical coordinates: changing basis $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{e}_z$.

Left: calculation of $f(r, \theta, z) := r^2 \cdot \sin(\theta) + 2z$ by 'hand'.

Right: 🐝 Check with the free CAS WEB REDUCE.

📄 Run EIGENMATH script

🐝 Run WEBREDUCE.

Exercise 6. Let $\mathbf{F}(x, y, z) = z\mathbf{e}_1 - 2x\mathbf{e}_2 + y\mathbf{e}_3 = (z, -2x, y)$.

- Write F in cylindrical coordinates.
- Then calculate $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ in *cylindrical coordinates*.
- Check your result with WEB REDUCE and the input

```
load_package avector;
coordinates r, theta, z;
scalefactors(1, r, 1);
Fr      := z*cos(theta) - 2*r*cos(theta)*sin(theta);
Ftheta  := -z*sin(theta)-2*r*cos(theta)^2;
Fz      := r*sin(theta);
F := avec(Fr,Ftheta,Fz);
div F;
curl F;
```

🐝 Run WEBREDUCE.

📄 Run EIGENMATH script

Exercise 7.

Derive formular's for $\nabla f, \nabla \cdot \mathbf{F}, \nabla \times \mathbf{F}, \Delta f$ in *parabolic coordinates*, i.e. $(x, y, z) = (\frac{1}{2}(u_1^2 - u_2^2), u_1 u_2, u_3)$, cf. [6, p.158, P.86].

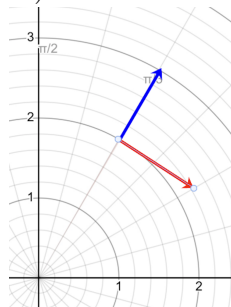
📄 Run EIGENMATH script

Exercise 8.

a. If $f(u_1, u_2, u_3) = u_1^2 u_3^3$ and $\mathbf{F} = u_3 \mathbf{e}_1 - u_2^2 \mathbf{e}_2 + 2u_1^2 u_2 \mathbf{e}_3$ find $\nabla f, \nabla \cdot \mathbf{F}, \nabla \times \mathbf{F}, \text{div}(f\mathbf{F}), \text{curl}(f\mathbf{F}), \Delta f$ in general 'curvilinear' coordinates..

Σ Run EIGENMATH script

b. Check a. by hand.

Exercise 9. (∇f in polar coordinates)

Polar coordinates: changing basis $\mathbf{e}_r, \mathbf{e}_\varphi$.

Get expressions for the operators $\nabla f, \nabla \cdot \mathbf{F}, \nabla \times \mathbf{F}, \Delta f$ in *polar coordinates* in $\mathbb{R}^2$.

Σ Run EIGENMATH script

Exercise 10. (Vector Operators: Grad, Div and Curl from 'jem' at cse.iitb.ac.in) Redo the worked examples and the exercises in §5 and §6 of graddivcurl.pdf.

Of special interest is the example Q2 with their solution.

Σ Run EIGENMATH script

Exercise 11. (tangential plane and gradient)

Do the exercises in ROOLFS: gradient.pdf using EIGENMATH.

Σ Run EIGENMATH script

Exercise 12. (vector operators in polar coordinates)

Study the text ROOLFS: Polarkoordinaten and control the results using EIGENMATH.

Σ Run EIGENMATH script

Exercise 13. (curl in spherical polar coordinates)

Find $\text{curl}(a)$ in (i) Cartesians and (ii) Spherical polars when $a = x(xI + yJ + zK)$, cf. graddivcurl.pdf.

Σ Run EIGENMATH script

Exercise 14. (Simple examples of $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$)

Read TONG [8, p.55 ff] and control the results using EIGENMATH.

Σ Run EIGENMATH script

Exercise 15. (Laplacian example by FLEISCH)

a. Study the example in FLEISCH [1, p.59 ff] VT and control the result using EIGENMATH.

b. Do the exercises 2.7 – 2.12 ibid.

Σ Run EIGENMATH script

Exercise 16. (Problem set in 'Introduction to Vector Calculus')

Solve a lot of the problem set Introduction to Vector Calculus and control/reproduce the results using EIGENMATH.

 Run EIGENMATH script

Exercise 17. (Problem set in 'Imath.osu.edu')

Study examples 1 to 5 in *Cylindrical and Spherical Coordinates* and control/reproduce the results using EIGENMATH.

 Run EIGENMATH script

Exercise 18. (Problem set of ROHINI College)

Google for 'rohini vector calculus' and choose the entry 'Unit-2 Vector Calculus.pdf'. Then study some examples and control/reproduce the results using EIGENMATH.

 Run EIGENMATH script

Exercise 19. (Divergence and Curl by HORAN/LAVELLE)

Do the Quiz in div-curl resp. div-curl and control/reproduce the results of the Quiz questions and the Examples using EIGENMATH.

 Run EIGENMATH script

Exercise 20. (A potpourri of worked examples)

 a solenoidal vector field EIGENMATH script

 unit normal vector to a surface EIGENMATH script

 curl of vector field $F = x \cdot (xI + yJ + zK)$ in Cartesian EIGENMATH script

 curl of vector field $F = x \cdot (xI + yJ + zK)$ in spherical EIGENMATH script

 divergence of $F = r * C$ in Cartesian and Spherical coord's EIGENMATH script

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